Shloke Binani

33A Canal Circular Road, Swarnamani, Kolkata-700054

+91 7003855628 | shlokebinani@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION

VELLORE INSTITUTE OF TECHNOLOGY, Vellore, India 2023 - 2027
Bachelor of Technology in Electronics and Communication Engineering **CGPA:** 8.68/10.0
Haryana Vidyamandir School, Kolkata | CBSE Class XII (2022 - 2023) **Score:** 80%
Calcutta Boys' School, Kolkata | ICSE Class X (2020 - 2021) **Score:** 93.2%

TECHNICAL SKILLS

- **Firmware & Embedded**: Embedded C, Assembly Language, 8/32-bit Microcontrollers (8051, ARM7), Serial Communication & I/O Interfacing, Compilers, Assemblers & Simulator Debuggers
- **Programming & Software Design**: C, C++, Python, MATLAB, Java (DSA), SQL, Object-Oriented Programming (OOP) & Design
- **Digital Logic & VLSI**: Digital Logic & Circuits, Verilog (Digital Design Lab), Cadence Virtuoso, Static & Dynamic CMOS Logic, Circuit Simulation, Transistor Sizing
- **Testing, Tools & Analysis**: Root Cause Analysis, Unit & Integration Testing, Design Documentation, Linux, Git, Docker, Statistical Analysis

ACADEMIC & HARDWARE PROJECTS

Manchester Carry Chain Adder | *Cadence Virtuoso, Digital Logic & Circuits, CMOS Simulation* Report

- Designed, documented, and simulated static CMOS and dynamic digital logic circuits in **Cadence Virtuoso**, evaluating static vs. dynamic trade-offs through systematic transistor sizing
- Analyzed and minimized critical path propagation delay across the carry chain, achieving an **18% speed improvement** over baseline CMOS adder implementations

Paradox: High-Concurrency Ticket Booking Platform | *React, Next.js, Node.js, FastAPI, PostgreSQL* Link

- Developed a real-time booking platform deployed on Neon, Vercel, and Render, implementing row-level database locking to guarantee transaction consistency under high concurrency

ResiliNet: Decentralized Network Routing Simulator | *Python, Communication Protocols, Fault Testing* GitHub

- Developed priority-aware packet routing algorithms across a **3-node** testbed and executed integration-level fault-injection testing to validate recovery paths, cutting peak latency by **89.8%** with **zero packet loss**

MS-HQNet: EMG Signal Classification Pipeline | *Python, Signal Processing, Algorithm Development*

- Developed **SM-SSA** and **NCA** signal processing and feature extraction algorithms paired with a hybrid **6-qubit QCNN-MLP** classifier across **600 electromyographic signals**, reaching **99.17% accuracy**

EXPERIENCE

Kofuku Technologies Pvt. Ltd., On-Site May 2025 - July 2025
Software Engineer Intern - Full-lifecycle software design, specification, testing, and debugging

- Reduced system failure rates by **30%** across distributed services by designing modular software interfaces and performing **root cause analysis** on authorization faults
- Cut reservation processing latency by **60%** by integrating real-time synchronization and automated speech-to-text processing pipelines
- Decreased QA regression cycles by **20%** by writing formal design specifications and establishing automated unit and integration testing suites
- Saved **8 to 12 hours per week** of manual cross-team coordination by developing event-driven workflow automation routines

LEADERSHIP

Society of Automotive Engineers (SAE-VIT) | *Marketing, Design & PR Head* May 2024 - May 2026

- Directed multi-channel outreach campaigns reaching **1,000+ students**, securing university administration buy-in and expanding chapter outreach
- Coordinated cross-functional engineering and operations teams across **40+ volunteers** and **12 major university events** during graVITas 25 and SAE AirShow 25, increasing participation by **30 to 45%** with 100% on-time execution

ACADEMIC ACHIEVEMENTS & CERTIFICATIONS

- Solved **250+ problems** across LeetCode & GeeksforGeeks (Easy to Hard), mastering Advanced Data Structures, Algorithms, and SQL
- Awarded the **Elite Category Chapter Award** for SAE-VIT for excellence in operations, 2025 to 2026
- Awarded the **Leadership Certificate** for exceptional contribution and management by Vellore Institute of Technology
- Won **Runner-Up** position at Central Hack for leveraging Web3 Crisis Protocols
- Won **Runner-Up** position at Providence Leap Ideathon for AI-Driven Healthcare Benefit Orchestration

Certifications: Pandas, Kaggle; Statistics OnRamp (2024) & MATLAB OnRamp (2024), The MathWorks, Inc.